ARYAN KUPEKAR

Electronics & Telecom Engineering Student | VLSI Physical Design, Verification & Embedded Hardware
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Professional Summary

Electronics & Telecommunication Engineering student at SPIT Mumbai seeking internships and full-time roles (from 2028) in VLSI physical design, design verification and embedded hardware. Built projects on FPGA, PIC18, ARM (LPC2148) and STM32, including a DIPEX 2025 nationally recognized project.

Education

B.Tech, Electronics & Telecommunication Engineering 2025 – Expected 2028
Sardar Patel Institute of Technology (SPIT) Mumbai, India
CGPA: 8.65/10 (up to Semester 5) | Coursework: Analog & Digital Electronics, Network Theory, Microcontrollers

Diploma, Electronics & Telecommunication Engineering 2022 – 2025
ICRE Gargoti
Aggregate: 95% | Best Academics Award 2025

Skills

Core Strengths: Digital Design | Analog Electronics | CMOS Fundamentals | Embedded Hardware | Debugging
Microcontrollers: PIC18, ARM LPC2148, STM32, ESP32, ESP8266, 8051
Hardware Design: Discrete Switching Circuits, Relay Control, Analog Signal Conditioning, Sensor Interfacing
Interfaces: UART, SPI, I2C, ADC
HDL & Firmware: Basic Verilog and VHDL (RTL coding), Basic Embedded C
Tools: Xilinx Vivado, Microwind, LTspice, Multisim, MATLAB, Proteus, Keil, STM32CubeIDE
Soft Skills: Communication, leadership (FETS Secretary), teamwork (3-member FPGA team), troubleshooting

Projects

Microcontroller-Based Cargo Vehicle Overload Protection System
STM32 (Blue Pill), Load Sensing, Relay, Buzzer GitHub

- Built an STM32-based system that monitors vehicle load, calibrated to the set overload limit.
- Designed relay and buzzer switching circuitry that alerts the driver and temporarily cuts the ignition on overload.

FPGA-Based Level Detection and Automated Load Management System
FPGA, VHDL/Verilog, Testbenches, IR and Ultrasonic Sensors LinkedIn Post

- Implemented an FPGA-based load management system in VHDL/Verilog and verified it with testbenches.
- Used two IR sensors for level detection, an ultrasonic sensor for obstacles, and a servo motor to even out loads.

Dumper Weight Measurement System DIPEX 2025 (National Level)
Load Cells, Microcontrollers GitHub

- Integrated load cells and microcontrollers into a dumper weight measurement and monitoring system.
- Earned national-level recognition at DIPEX 2025.

Experience

Secretary Aug 2025 – May 2026
Forum of Electronics and Telecommunication Students (FETS), SPIT Mumbai, India

- Led communication and coordination for the department's student forum.

Technical Intern May 2025 – Jul 2025
Classic Communication

- Troubleshot PCBs, debugged firmware faults and validated electronic equipment.
- Tested circuits and performed preventive maintenance on electronic systems.

Achievements & Activities

- National-level recognition at DIPEX 2025 for a hardware-based dumper weight measurement project.
- Finalist, BugBuster and Unplugged hardware hackathons (2026).
- Best Academics Award 2025 with 95% aggregate in Diploma.
- Competitive football player; developed teamwork and leadership skills.